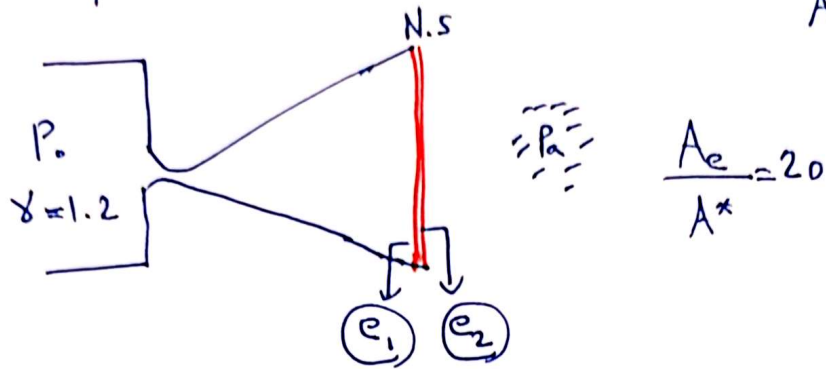


Friday - Feb-14-2025: Nozzle examples & Discussions

Example 1: A C-D nozzle has $\frac{A_e}{A^*} = 20$, find $\left(\frac{P_0}{P_a}\right)_{crit}$



Solution:

First: At (e_1) , the flow is supersonic & M_{e_1} is determined:

$$\begin{cases} \frac{A_e}{A^*} = 20 \\ \gamma = 1.2 \end{cases} \xrightarrow{\text{online calculator}} M_{e_1} = 3.76$$

Second: $P_{e_1} = \left(\frac{P_{e_1}}{P_0}\right) \times P_0 = 0.0051 \times P_0 = \frac{P_0}{197.24}$

online calculator

Third: $\frac{P_{e_2}}{P_{e_1}} \begin{cases} M_{e_1} = 3.76 \\ \gamma = 1.2 \end{cases} \xrightarrow[\text{for N.S.}]{\text{online calculator}} \frac{P_{e_2}}{P_{e_1}} = 15.33$

Forth: $P_{e_2} = 15.33 P_{e_1} = 15.33 \times \frac{P_0}{197.24} = 0.0775 P_0$

$$P_{e_2} = P_a \implies \frac{P_0}{P_{e_2}} = \frac{P_0}{P_a} = \frac{1}{0.0775} = 12.90 = \left(\frac{P_0}{P_a}\right)_{crit}$$

Note: $\frac{A_e}{A^*}$ & γ determines $\left(\frac{P_0}{P_a}\right)_{crit}$ (P-1)

Example 2: In the previous question, $p_o = 1 \text{ MPa}$. Will the nozzle be shock free at 10 km altitude? Find M_e & p_e at this condition. ($\gamma = 1.2$, $\frac{A}{A^*} = 20$)

Solution: At 10 km altitude, $p_a = 26.5 \text{ kPa}$, $p_o = 1000 \text{ kPa}$

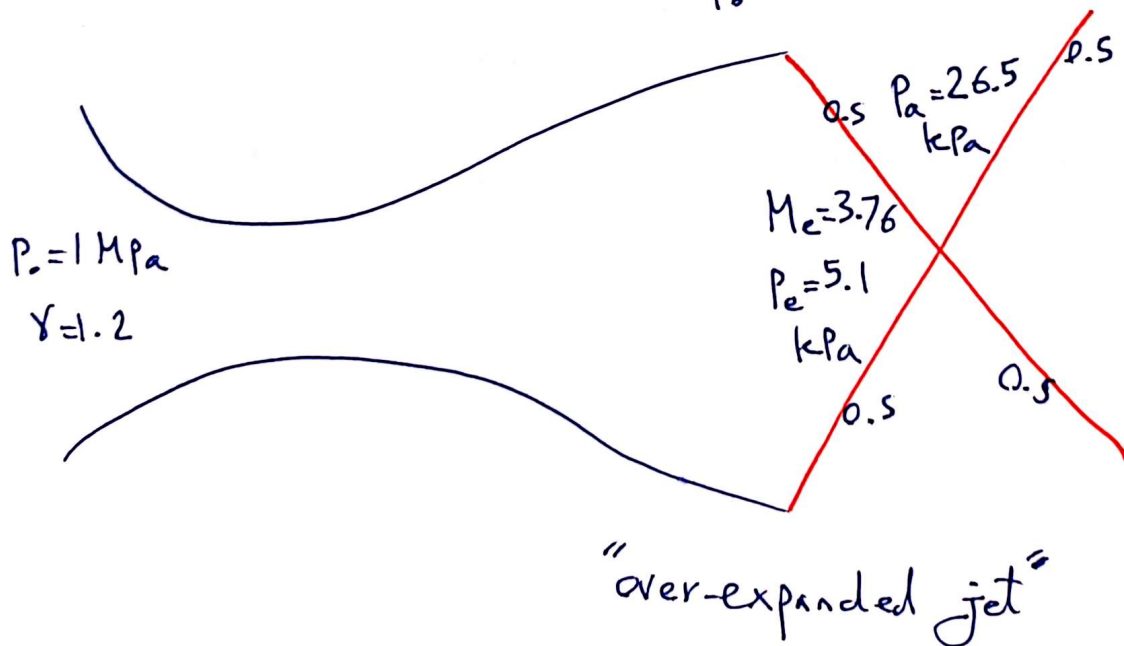
$$\left(\frac{p_o}{p_a} \right)_{\text{actual}} = \frac{1000}{26.5} = \boxed{37.74}$$

From previous example: $\left(\frac{p_o}{p_a} \right)_{\text{crit}} = 12.9$

$$\left(\frac{p_o}{p_a} \right)_{\text{actual}} > \left(\frac{p_o}{p_a} \right)_{\text{crit}} \longrightarrow \text{shock free nozzle}$$

$M_e = ?$ $\begin{cases} \frac{A_e}{A^*} = 20 \\ \gamma = 1.2 \end{cases} \longrightarrow M_e = 3.76 \text{ (as before)}$

$p_e = ?$ ($p_e \neq p_a$) $p_e = \frac{p_e}{p_o} \times p_o = 0.0051 \times 1000 \text{ kPa} = \boxed{5.1 \text{ kPa}}$



(P-2)

Example 3: what is the lowest altitude where the previous nozzle works shock-free?

Solution: $\frac{P_0}{P_a} \Big|_{\text{crit}}$ is the threshold of shock-free nozzle.

$$\frac{P_0}{P_a} \Big|_{\text{crit}} = 12.9, P_0 = 1000 \text{ kPa} \rightarrow P_a \Big|_{\text{crit}} = \frac{1000}{12.9} = 77.519 \text{ kPa}$$

Run MATLAB or ask chatGPT the critical altitude corresponding to this pressure $\Rightarrow h_{\text{crit}} = 2.2 \text{ km}$

If $h < h_{\text{crit}} \rightarrow$ shock is inside nozzle.

" $h = h_{\text{crit}} \rightarrow$ " " at exit plane of nozzle

" $h_{\text{crit}} < h < h_{\text{perfect-expanded}} \rightarrow$ oblique shock outside nozzle (over-expanded)

" $h = h_{\text{perfect-expanded}} \rightarrow$ No shock (Design condition)

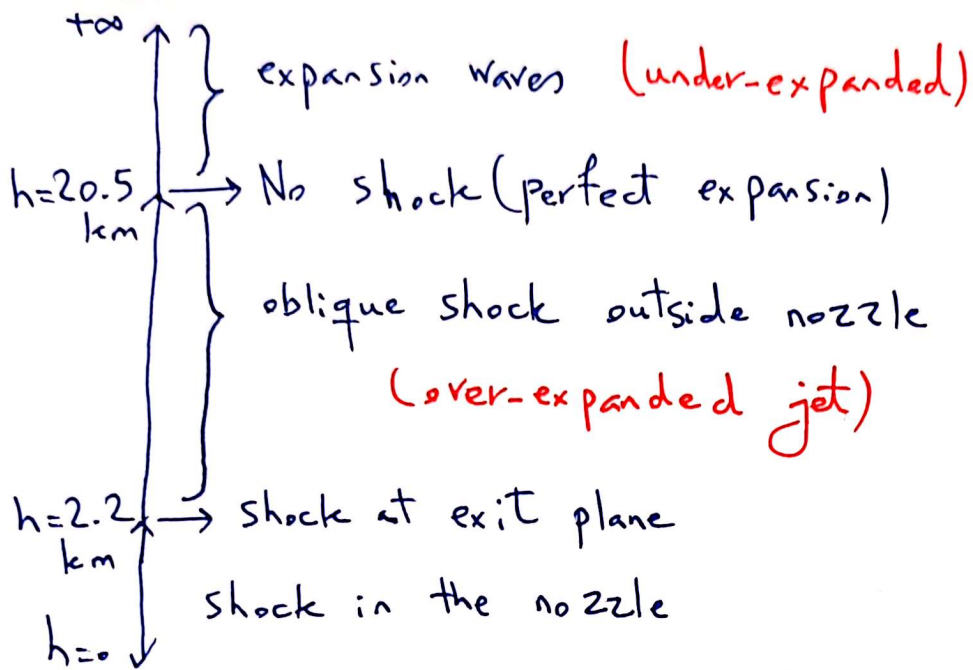
" $h > h_{\text{perfect-expanded}} \rightarrow$ Expansion waves (under-expanded)

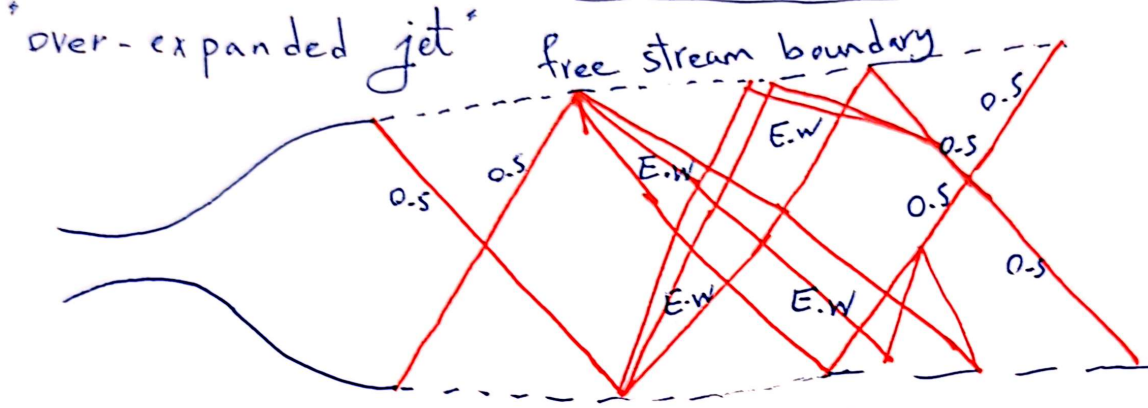
Example 4: Find $h_{\text{perfect-expanded}}$ for the previous nozzle.

Solution: At perfect expansion, $P_a = P_e)_{\text{supersonic}} = 5.1 \text{ kPa}$

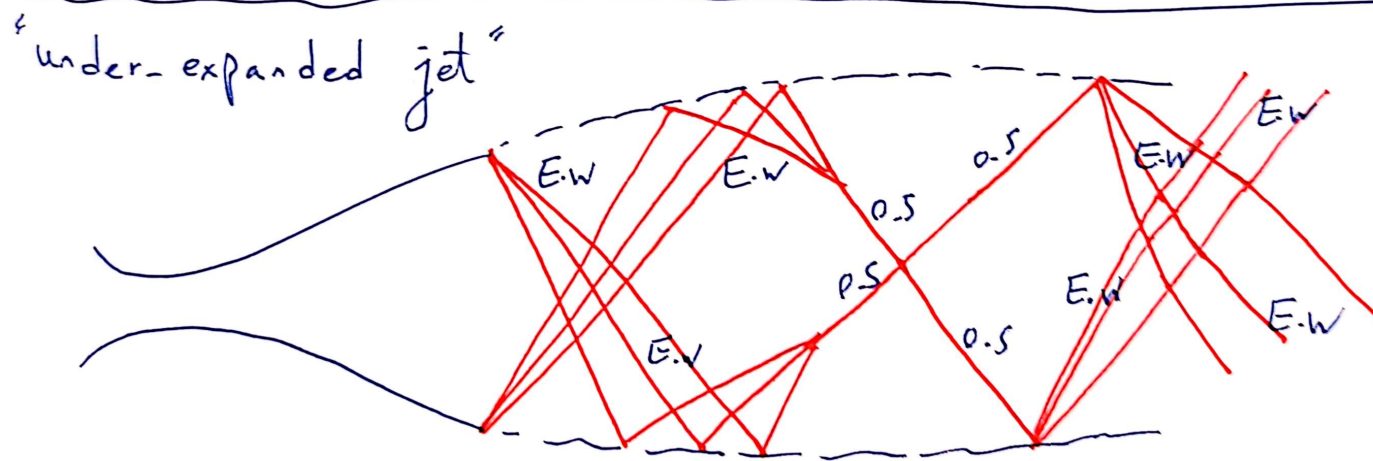
$$h(P = 5.1 \text{ kPa}) = 20.5 \text{ km}$$

(P-3)





Note: oblique shocks (o.s) reflect back from free surface as expansion wave (E.W) & vice versa.



Note: Expansion waves reflect back from free-surface as oblique shocks.

(P-5)